



STEAM®

The Price of Play: What Makes a Steam Game Financially Successful?

What Drives Financial Success on Steam?

DATA 485 · CAPSTONE PROJECT

SPRING 2026

Meet the Team



Data Analytics

Dimension 1 (Publisher Class) · Random Forest ·
Introduction

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Data Analytics

Dimension 2 (Price Tier) · Regression Analysis ·
Recommendations



Data Analytics

Dimension 3 (Review Score) · Data Prep &
Transformation

AGENDA

01 Introduction & Data · 02 Dimension Analyses · 03 Predictive Modeling · 04 Recommendations

Group 5 · Professor Winnie Li · DATA 485 — Data Analytics Capstone



Steam is the ultimate destination for playing, discussing, and creating games.

● ONLINE

26,265,818

● PLAYING NOW

6,667,951

01.

Introduction: The Price of Play

The video game industry has grown into one of the most profitable sectors of digital entertainment — a global business shaped by software development, consumer behavior, pricing strategies, reviews, and long-term player engagement. Understanding what drives revenue has become increasingly valuable for developers, publishers, investors, and platform stakeholders.



Steam is one of the largest and most influential PC gaming storefronts — an ecosystem for discovery, recommendations, and commercial success. Steam performance can strongly influence visibility, audience reach, and financial outcomes for PC titles.

Research Focus & Business Question

The Core Question

What factors are most associated with financial success among top-performing Steam games?

Rather than asking *which* games earned the most, this project explores *why* certain games outperform others.

Factors Examined

- Publisher classification (AAA, AA, Indie)
- Price tier and pricing strategy
- Review score performance
- Player engagement (average playtime)
- Copies sold and market share

This capstone project uses Gamalytic Steam analytics data to examine revenue performance patterns across 10,638 games. The project uses data cleaning, descriptive statistics, regression, Random Forest modeling, and dashboard development to identify key revenue drivers.

Data Description & Scope

Data Source

Gamalytic Steam Analytics API
Steam game performance dataset with estimated revenue, sales, player, pricing, review, and wishlist data

Dataset Size

10,568 games analyzed after cleaning, spanning multiple genres and publisher types

Key Variables

Revenue, price, avg. playtime, review score, review count, wishlists, publisher class, price tier, review tier

 **Scope & Limitations:** This analysis uses Gamalytic Steam analytics data, including estimated revenue, sales, pricing, review, wishlist, and playtime fields. Results should be interpreted as market-performance estimates, not official Steam financial records. The analysis identifies patterns across publisher, pricing, review, and engagement dimensions, but does not include outside factors such as marketing budget, franchise strength, streamer exposure, or platform visibility.

Data Cleaning & Transformation

The dataset contained 57 games with a missing value, which was dropped along with games that have "Hobbyist" publishers, leaving 10,568 rows of observations. Five variables exhibited moderate to substantial skewness. Log10, square root, and reflected square root transformations were applied to normalize distributions, successfully reducing skewness severely, validating their use in subsequent statistical testing.

Variable	Skew Issue	Transformation	Skewness (Before)	Skewness (After)
copiesSold	Substantially positive	$\log_{10}(\text{data\$copiesSold})$	13.2776	0.8336
revenue	Substantially positive	$\log_{10}(\text{data\$revenue})$	13.5153	0.4741
avgPlaytime	Substantially positive, one zero	$\log_{10}(\text{data\$avgPlaytime} + 1)$	1.5631	0.6901
price	Moderately positive, zeros	$\sqrt{\text{data\$price}}$	1.5751	0.5504
reviewScore	Moderately negative	$\sqrt{k - \text{data\$reviewScore}}$	-1.3928	0.5294



02.

Dimension 1

Publisher Class & Revenue Performance

AAA vs. AA vs. Indie

Does publisher scale determine financial success on Steam?

Publisher Class: Dataset Composition

Volume vs. Revenue — A Structural Disconnect

Indie games dominate the Steam marketplace by sheer count, yet AAA and AA titles capture the vast majority of revenue. The filtered dataset included 10,580 games total across three publisher classes.

70.3%

Indie Share

7,440 titles in dataset

20.0%

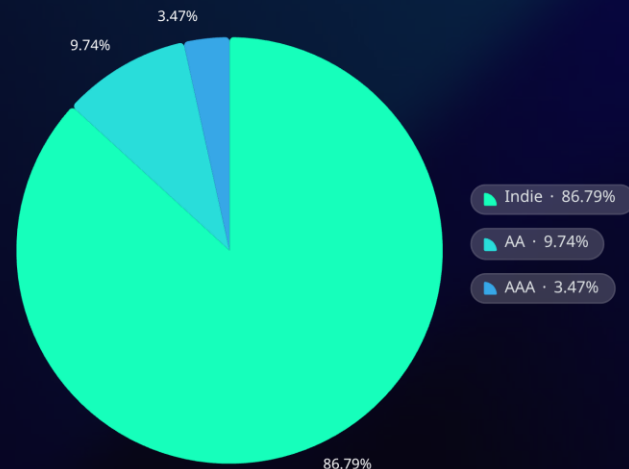
AA Share

2,112 titles in dataset

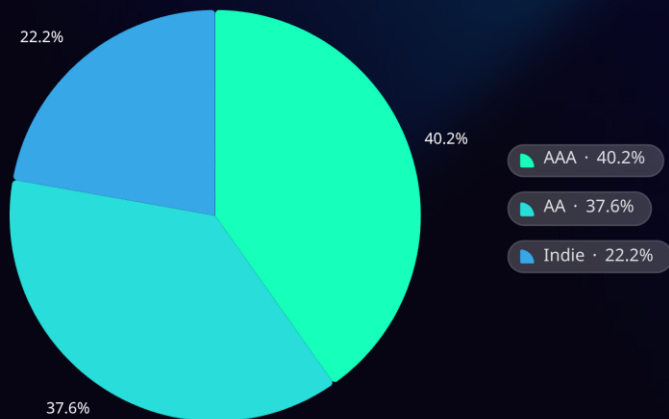
9.7%

AAA Share

1,028 titles in dataset



Publisher Class: Revenue Share



Revenue Tells a Different Story

Despite representing the smallest share of titles, AAA games captured **44.3% of total revenue (~\$27.2B)**. AA games followed at **36.2% (~\$22.2B)**. Indie games, despite comprising 70.3% of titles, earned just **19.5% (~\$12.0B)**.

Median revenue reinforces this gap:

- **AAA:** ~\$5.1M median revenue
- **AA:** ~\$1.4M median revenue
- **Indie:** ~\$407.2K median revenue



The typical AAA game earned over **12× more** than the typical Indie game.

Publisher Class: Statistical Findings

Welch ANOVA ($F = 1095.061, p < 0.001$)

Publisher class had a statistically significant relationship with mean log revenue. Because the groups had unequal variances, Welch ANOVA was used as the main test, and all pairwise comparisons were significant.

Adjusted Regression Support

Publisher class remained meaningful after controlling for price, review score, and average playtime. AA and Indie games showed lower log revenue than AAA games, suggesting publisher class captures market advantage beyond basic performance metrics.

Correlation Matrix Highlights

Sqrt price had the strongest relationship with log revenue, followed by log average playtime. Review score had a much weaker relationship, showing that pricing power and engagement were stronger revenue signals than review quality alone.

Key Takeaway

Publisher class is a real revenue divider in the Steam marketplace. Indie games lead in title count, but AAA leads in revenue power, while AA sits between the two with stronger performance than Indie but below AAA.

Categories ▾ Hardware ▾ Ways to Play ▾ Special Sections ▾

s Top Sellers Popular Upcoming Specials Trending Free

	Counter-Strike 2 FPS, Shooter, Multiplayer, Competitive Released: Aug 21, 2012	Free
	Wuthering Waves Open World, Anime, Free to Play, Action Released: Apr 28, 2025	Free
	Forza Horizon 6 Racing, Open World, Driving, Multiplayer Released: May 18, 2026	\$69.99
	Steam Deck Released: Jan 17, 2025	\$789.00
	Solarpunk™ Open World Survival Craft, Co-op, Base Building, Crafting Released: Jun 8, 2026	\$22.99
	Path of Exile 2 Action RPG, Hack and Slash, RPG, Isometric	

 Manage Downloads

02. Dimension 2 Price Tier

Under \$20 · \$20-\$40 · \$40+

Which pricing strategy generates the most revenue on Steam?

Price Tier: Descriptive Statistics

Revenue figures shown in actual dollars across three price tiers in the expanded dataset.

Price Tier	Count	% of Dataset	Mean Revenue	Median Revenue	Avg. Revenue Score
Under \$20	7,985	75.1%	\$2,547,191	\$392,892	5.67
\$20-\$40	2,132	20.0%	\$10,723,926	\$2,056,084	6.40
\$40+	521	4.9%	\$35,297,188	\$7,354,687	6.93

 The Under \$20 tier makes up ~75% of all games yet earns the **lowest average revenue**. The \$40+ tier earns the highest average revenue by a wide margin, despite representing under 5% of all games.

Price Tier: Revenue by Tier

Average Log Revenue by Price Tier



Stepwise Revenue Increase

Average revenue rises steadily from Under \$20 through to \$40+. The \$40+ tier shows the highest average revenue overall, though the gap between tiers is more incremental on a log scale than it first appears.

Premium pricing is associated with higher revenue, but the relationship is gradual rather than a sharp jump at any single tier.

✓ \$40+ games average **\$35.3M** in mean revenue, nearly 14× the Under \$20 mean of \$2.55M.

Price Tier: Statistical Testing

How Strongly Each Factor Relates to Revenue
(Only showing the three most relevant rows in this dimension)

	Revenue	Copies Sold	Price	Avg. Playtime	Review Score
Revenue	1.0000	0.8758	0.6216	0.5017	-0.1005
Price	0.6216	0.2672	1.0000	0.4851	0.0977
Avg. Playtime	0.5017	0.3392	0.4851	1.0000	-0.0385

Price Tier Revenue Comparison (Statistical Test)

Comparison	Difference	Significant?
\$20-\$40 vs Under \$20	0.726	Yes (p ≈ 0)
\$40+ vs Under \$20	1.256	Yes (p ≈ 0)
\$40+ vs \$20-\$40	0.530	Yes (p ≈ 0)

All 3 pairwise tier comparisons showed statistically significant revenue differences (p ≈ 0), confirming that each price tier generates meaningfully different revenue outcomes.

Counter-Strike 2



All Games > Free To Play Games > Dota Franchise > Dota 2

Dota 2



Price Tier: Key Takeaway

\$40+ Wins

Premium pricing is most strongly associated with higher net revenue. \$40+ games have both the highest average earnings and the widest range of revenue outcomes.

Price Drives Revenue

Price and revenue show a strong positive relationship ($r = 0.62$) across the expanded dataset, confirming that pricing strategy is one of the clearest drivers of revenue on Steam.

Under \$20 Underperforms

Despite dominating by volume (75%), the budget tier consistently underperforms all others. Low price alone is the least effective strategy.

02.

Dimension 3

Review Score & Revenue Performance

High vs. Low Review Score

Does critical reception translate to commercial success on Steam?

Community Hub



For over two decades, Counter-Strike has offered an elite competitive experience, one shaped by millions of players from across the globe. And now the next chapter in the CS story is about to begin. This is Counter-Strike 2.

RECENT REVIEWS: [Very Positive](#) (71,707)
ENGLISH REVIEWS: [Very Positive](#) (2,551,036)
RELEASE DATE: Aug 21, 2012
DEVELOPER: [Valve](#)
PUBLISHER: [Valve](#)

Popular user-defined tags for this product:

[FPS](#) [Shooter](#) [Multiplayer](#) [Competitive](#) [Action](#) [eSports](#) [+](#)

Review Score Tier: Descriptive Statistics

All statistical testing was conducted on log-transformed revenue. Raw figures are reported alongside for interpretability. The Overwhelmingly Negative (<20%) and Mostly Negative (20–39%) tiers contain only 6 and 74 games respectively, meaning that they have too few observations to draw reliable conclusions.

Review Score Tier	Count	% of Dataset	Mean Revenue	Median Revenue	Std Dev Revenue	Min Revenue	Max Revenue	Mean Log Rev.
Overwhelmingly Positive (≥95%)	1,543	14.6%	\$9.50M	\$649,204	\$38.30M	\$5,550	\$763.4M	5.94
Very Positive (80–94%)	4,191	39.7%	\$6.19M	\$671,959	\$28.10M	\$6,444	\$729.3M	5.93
Mostly Positive (70–79%)	3,466	32.8%	\$4.51M	\$584,481	\$19.40M	\$5,534	\$435.9M	5.85
Mixed (40–69%)	1,288	12.2%	\$3.82M	\$448,736	\$20.86M	\$2,657	\$541.3M	5.72
Mostly Negative (20–39%)	74	0.7%	\$2.44M	\$335,443	\$8.68M	\$5,670	\$66.7M	5.48
Overwhelmingly Negative (<20%)	6	0.1%	\$836,671	\$740,068	\$697,847	\$66,441	\$1.68M	5.71

 The Very Positive tier dominates by volume (~39.7%) yet the Overwhelmingly Positive tier leads in mean log revenue (5.94). The substantial mean-to-median gap in the Overwhelmingly Positive tier (\$9.50M mean vs. \$649,204 median) confirms its elevated mean is driven by blockbuster outliers rather than broad success. Notably, the Mixed (5.72), Mostly Positive (5.85), and Very Positive (5.93) tiers show a more gradual log revenue spread than previously observed, though the practical differences remain modest relative to their review score gaps. Only reaching Overwhelmingly Positive (≥95%) produces a sharp and clear revenue separation from all other tiers.

Review Score Tier: Revenue by Tier



Small Stepwise Increase

Mean log revenue rises gradually across tiers but does not follow a clean stepwise pattern. The Overwhelmingly Negative (<20%) and Mostly Negative (20–39%) tiers contain only 6 and 74 games respectively, too few observations to draw reliable conclusions.

Moving up from Mixed (5.72) to Mostly Positive (5.85) to Very Positive (5.93), and then finally to Overwhelmingly Positive (5.94) shows a consistent but small stepwise increase, with the smallest gap occurring at the very top. Gains in mean log revenue diminish as review scores rise, suggesting that moving from Mixed to Very Positive yields more marginal benefit than the review score jumps might imply.

- ✓ Overwhelmingly Positive games average \$9.50M in mean revenue, nearly 2.5× the Mixed tier mean of \$3.82M. **However, the stark gap between the Overwhelmingly Positive tier's mean (\$9.50M) and its median (\$649,204) signals that a small number of blockbuster outliers are pulling the average upward, rather than a broad revenue lift across all games in the category.**

Review Score Tier: Statistical Testing

Correlation Matrix & Tukey Post-Hoc

	Revenue	Review Score	Avg. Playtime	Price
Revenue	1.000	-0.1005	0.5017	0.6216
Review Score	-0.1005	1.000	0.636	0.0977

Review Score shows negligible correlations across all variables, nearly zero with Revenue ($r = -0.1005$), Avg. Playtime ($r = -0.0385$), and Price ($r = 0.0977$). This suggests that review score is largely independent of the other predictors and has minimal linear association with revenue, challenging the assumption that having more positive reviews equal more revenue.

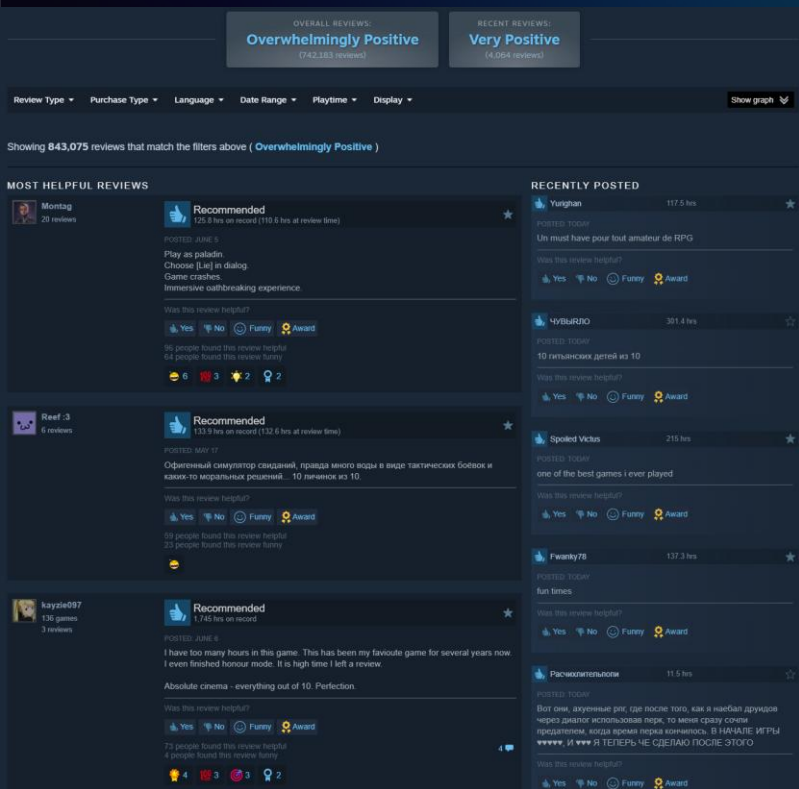
Comparison	Diff	Lower	Upper	p adj
Very Positive – Ovwh. Positive	-0.0096	-0.0773	0.0580	0.9986
Mostly Positive – Ovwh. Positive	-0.0879	-0.1574	-0.0183	0.0043 **
Mixed – Ovwh. Positive	-0.2244	-0.3102	-0.1387	< 0.0001 *
Mostly Negative – Ovwh. Positive	-0.4590	-0.7294	-0.1885	< 0.0001 *
Ovwh. Negative – Ovwh. Positive	-0.2323	-1.1619	0.6972	0.9805
Mostly Positive – Very Positive	-0.0783	-0.1304	-0.0261	0.0003 *

Review Score Tier: Statistical Testing Cont.

Tukey Post-Hoc

Comparison	Diff	Lower	Upper	p adj
Mixed – Very Positive	-0.2148	-0.2872	-0.1424	< 0.0001 *
Mostly Negative – Very Positive	-0.4494	-0.7159	-0.1829	< 0.0001 *
Ovwh. Negative – Very Positive	-0.2227	-1.1511	0.7057	0.9838
Mixed – Mostly Positive	-0.1366	-0.2107	-0.0624	< 0.0001 *
Mostly Negative – Mostly Positive	-0.3711	-0.6381	-0.1041	0.0011 **
Ovwh. Negative – Mostly Positive	-0.1445	-1.0730	0.7841	0.9978
Mostly Negative – Mixed	-0.2345	-0.5062	0.0371	0.1360
Ovwh. Negative – Mixed	-0.0079	-0.9378	0.9220	1.0000
Ovwh. Negative – Mostly Negative	0.2266	-0.7380	1.1913	0.9852

Tukey HSD post-hoc testing found 9 of 15 pairwise comparisons statistically significant. The largest gap was between Mostly Negative and Overwhelmingly Positive (diff = -0.459, $p < 0.001$), with lower-scoring tiers consistently showing reduced log revenue. Overwhelmingly Negative returned no significant comparisons, likely due to its small sample size ($n = 6$) inflating confidence intervals rather than a true absence of effect.



Review Score Tier: Key Takeaway

Review Scores Alone Don't Drive Revenue

Review score has a near-zero correlation with revenue ($r = -0.10$), meaning high review score do not reliably predict higher earnings. Most revenue differentiation comes from price and playtime, not review score.

Only the Top Tier Stands Out

Overwhelmingly Positive games average \$9.50M in mean revenue, but this is driven by blockbuster outliers as the median is just \$649,204. Reaching $\geq 95\%$ positive reviews correlates with higher revenue, but does not guarantee it.

From Mixed to Very Positive: Little Revenue Difference

Mixed, Mostly Positive, and Very Positive tiers show only modest differences in mean log revenue (5.72 $\rightarrow$ 5.85 $\rightarrow$ 5.93). Achieving a "good" review score provides limited revenue advantage over a "mixed" one.

03. Predictive Modeling Regression Analysis

Using data to identify which game characteristics most strongly predict Steam revenue, and by how much

TOP SELLERS

Top 100 selling games right now, by revenue

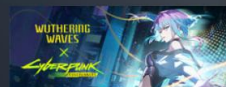
RANK

1



Counter-Strike 2

2



Wuthering Waves

3



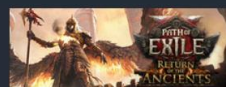
Forza Horizon 6

4



Steam Deck

5



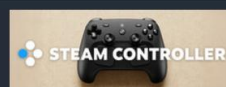
Path of Exile 2

6



Solarpunk™

7



Steam Controller

8



007 First Light

Regression: Model & Results

Model Setup

- Built a predictive model to estimate game revenue based on measurable game characteristics
- Inputs included price, review score, average playtime, and publisher class (AAA, AA, or Indie)
- All publisher comparisons are made relative to AAA games as the benchmark

Key Regression Findings

- Average playtime was the strongest positive predictor of revenue
- Price was also a meaningful positive driver of revenue
- Review score was not significant after controlling for the other variables
- Indie games earned significantly less than AAA games, even when price and playtime were the same

 This model tells us which factors matter and by how much, laying the groundwork for the deeper pattern analysis that follows.

03.

Predictive Modeling

Random Forest

A Random Forest regression model to capture nonlinear patterns and complex variable interactions in Steam revenue prediction



Random Forest: Model Setup & Performance

Model Configuration

- Target:** log_revenue
- Log-transformed to reduce skew from extreme high-earning titles.
- Predictors:** publisherClass, price_tier, review_tier, price, reviewScore, avgPlaytime, review_count, wishlists
- Excluded:** Hobbyist titles, revenue, copiesSold, log_copiesSold, players_total, total_revenue, M1/Y1 revenue and sales, yearly revenue/sales fields
- Observations:** 10,572 complete AAA, AA, and Indie observations
- Split:** 80% training / 20% testing, stratified by publisher class
- 8,487 training observations / 2,085 testing observations
- Trees:** 500 trees, mtry = 2

Model Performance

0.907

R-Squared

Explained 90.7% of variation in log revenue on test set

0.242

RMSE

Root Mean Squared Error on test set

0.189

MAE

Mean Absolute Error on test set

- ✔ Visual diagnostics confirm strong fit — the Actual vs. Predicted plot shows most points clustering tightly around the diagonal reference line. The model should be interpreted as predictive, not causal.

Random Forest: Feature Importance



What the Model Tells Us

Review count is the dominant predictor — market attention and player response volume are the strongest signals of revenue performance. Games that generate more review activity tend to have stronger revenue outcomes.

Price, wishlists, and average playtime are the next strongest predictors, showing that revenue is shaped by a combination of pricing power, pre-purchase demand, and player engagement.

Review score contributed to the model but ranked below review count, price, wishlists, and playtime. Volume of player response is more predictive than score quality alone.

Publisher class, price tier, and review tier were still useful but ranked lower than direct market-demand and engagement variables.

❏ In this updated model, categorical fields are reported at the variable level (publisherClass, price_tier, review_tier) rather than as separate dummy categories.

Regression vs. Random Forest: Complementary Lenses

Regression Analysis

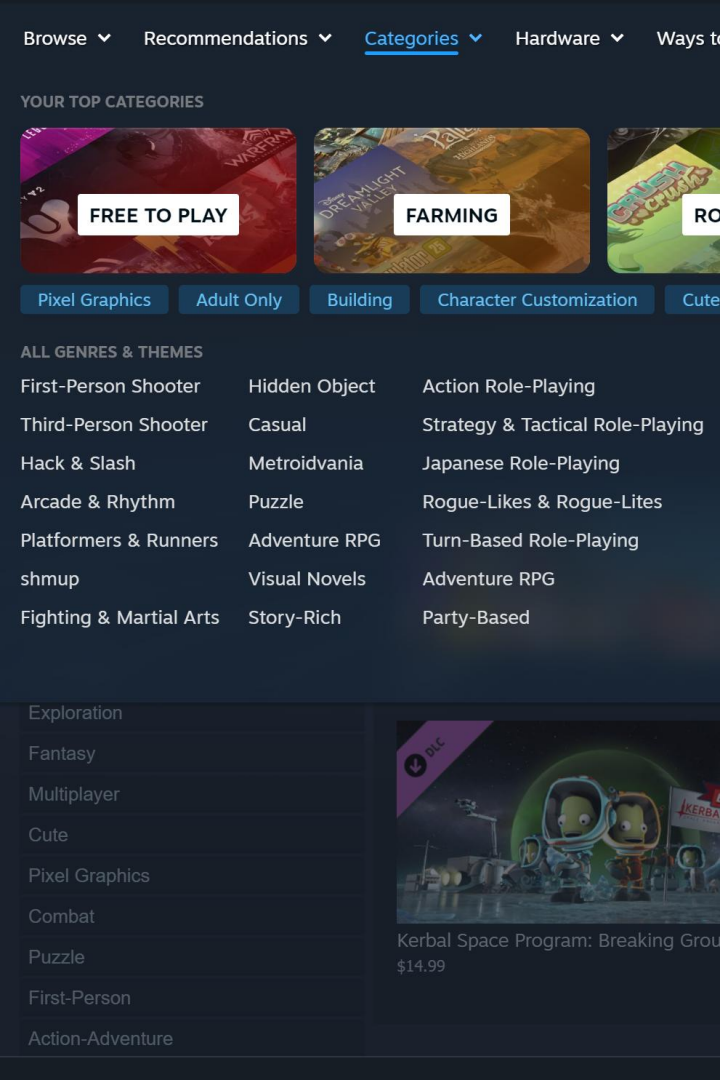
Establishes direct, structural relationships between variables.
Quantifies the specific effect of each predictor on log revenue.
Confirms Indie games underperform AAA/AA even after controlling for other factors.

Random Forest Model

Captures nonlinear patterns and complex variable interactions without pre-specifying relationships. Provides definitive feature importance ranking. In the updated whole-dataset model, `review_count` was the strongest predictor, followed by price, wishlists, and `avgPlaytime` — confirming that demand signals and engagement drive revenue more than review score alone.

- ✓ The Random Forest does not replace the regression — it complements it. Together, they show both **what** drives revenue and **how strongly** each factor contributes when all dimensions interact simultaneously.

Visual diagnostics (Actual vs. Predicted plot, Residual plot) confirm the model captures broad structural patterns with strong fit ($R^2 = 0.894$). A few games show larger prediction errors at the higher revenue end, suggesting additional variables like genre, marketing spend, and Steam visibility could improve future model performance.



04. Recommendations

From analysis to action: What the data tells us about building successful games on Steam

Recommendations



Prioritize Engagement Above Everything

Before optimizing price or chasing reviews, build features that keep players coming back. Playtime is the strongest predictor of revenue, replay-ability, live service content, and multiplayer are your highest-ROI investments.



Indie Developers: Win on Community

Publisher scale creates a revenue gap that price and quality alone cannot close. Indie studios should focus resources on building loyal, engaged communities the one area where they can compete regardless of budget.



Price Strategically

Premium pricing (\$40+) maximizes revenue, but only if your game has AAA-level production quality and marketing reach to back it up. For indie developers without that backing, mid-range pricing (\$20-\$40) still significantly outperforms budget pricing.



Don't Chase Review Scores Alone

Review score did not significantly predict revenue among games. A good game matters, but directing resources toward engagement features will do more for revenue than optimizing for critical reception.



Limitations Reminder: Findings reflect patterns among the 10,568 games analyzed in this dataset and may not generalize to the broader Steam library of 50,000+ titles. Also, free-to-play games were left out as they have massive outliers in revenue.

Q&A

Thank You

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